



Development of a Tool for Assessing the Quality of Health Services in Municipalities: First Insights

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Abstract. This article elaborates on the development of an innovative tool designed to assess the quality of health services within municipalities. Through the introduction of a user-friendly questionnaire application accessed via QR code scanning, efficient and accessible data collection is facilitated. Furthermore, for the analysis and interpretation of gathered data, a dashboard application linked to Microsoft's Power BI software has been integrated, offering dynamic and comprehensive visualization of responses. The forthcoming testing phase within a real-world environment aims to showcase the feasibility and efficacy of the system. Preliminary results highlight the practical value of the tool, providing valuable insights for the ongoing enhancement of municipal health services. This project represents a noteworthy advancement in the field, merging technological innovation with the evaluation of health services, thus presenting a substantial contribution. It serves as a testament to the intersection of technology and healthcare, demonstrating the potential for impactful improvements in service quality and delivery.

Keywords: Health Units Assessment Tool · Visual Studio · Power BI · User Satisfaction · User Interface

1 Introduction

Currently, companies in the healthcare sector are competing with each other to maximize the provision of services, such as public service, average waiting times, precision in tasks performed, among others. Customer/patient satisfaction is increasingly a crucial point regarding business growth, and there has been a growing and continuous investment in specific systems capable of intelligently and effectively evaluating the satisfaction level of the entire sample involved (users and professionals).

In Portugal, the National Health Service is the main entity providing essential health services, which include: hospitals, medical posts, health centers and USF (family health unit).

According to the official website of the National Health Service [1], this set of institutions and services, dependent on the Ministry of Health, has the mission of ensuring

access to healthcare for all citizens within the limits of available human, technical, and financial resources.

The health centers and hospitals in Portugal have a significantly outdated evaluation system compared to well-known companies, such as the widely recognized Swedish group IKEA. Typically, these establishments rely on the traditional “suggestion box,” usually placed near the respective building’s reception. However, this approach not only fails to encourage or engage users but also does not contribute to an improvement in healthcare units.

The project’s objective revolves around developing a tool for evaluating patient satisfaction with the quality of healthcare services. More specifically, the goal is to replace the suggestion boxes with a digital evaluation tool that aims to address the current weaknesses in the service. This approach offers insights to implement corrective measures and helps establish performance goals for the future.

1.1 Goals

The National Health Service (SNS) requires the implementation of quality assessment methodologies and provides its clients with tools to measure user satisfaction with service quality. The current instrument used in healthcare units is a questionnaire filled out on paper at the unit itself. This master’s dissertation project emerges within the scope of reconfiguring this tool to make the current procedures more transparent.

The primary objective of the project is to create a Healthcare Services Evaluation Tool for Municipalities, capable of assessing the implementation of a motivation strategy for questionnaire completion. The development of this project will also be complemented by the following secondary objectives:

- Eliminate the use of paper;
- Improve the current healthcare unit assessment system;
- Accessibility to the platform from anywhere;
- Design a simple and accessible questionnaire for the illiterate population (approximately 5.2% according to the 2011 census data);
- Zoom functionality in the questionnaire;
- Implementation of a Dashboard presenting relevant static data;
- Design a simple and appealing questionnaire to encourage its completion.

1.2 Research Hypotheses

The development of a Health Services Evaluation Tool for Municipalities, capable of assessing the implementation of a motivation strategy for questionnaire completion, aims to address the following research hypotheses, which involve studying:

1. Is it feasible to develop a system capable of collecting more questionnaires than the current one?
2. Can municipal technicians comprehend the progress based on the developed dashboards?
3. Is it possible to motivate residents to complete the questionnaires in full?

1.3 Organization

This paper is organized in 4 sections. In Sect. 2, the adopted research methods are detailed, providing a clear insight into the strategies and approaches used to achieve the study's objectives. The practical implementation of the project is described in Sect. 3, meticulously highlighting the steps and decisions made during the development of the health services evaluation tool. Finally, Sect. 4 presents a report on the current state of the project, showcasing milestones achieved, challenges faced, and outlining the planned next steps. This structure provides a holistic understanding of the development, methods, and ongoing progress of this innovative project.

1.4 Weaknesses of the Current System

The current system exhibits notable weaknesses. The use of paper questionnaires raises concerns, as these documents can be lost, resulting in the loss of valuable responses. Furthermore, this approach is not environmentally sustainable and requires significant manual labor for analysis.

The lack of automation in data analysis is another critical weakness. The absence of this feature can lead to long waiting periods for obtaining results, delaying the implementation of corrective measures to enhance services. This delay not only hampers operational efficiency but also incurs substantial costs for institutions, both in terms of hours spent on manual analysis and materials used, such as paper and ink cartridges. Therefore, the need for a transition to more efficient and sustainable methods becomes evident.

1.5 State of the Art

Currently, businesses are cognizant of the significance of their services and products, and as a result, they strive to provide the best possible experience to distinguish themselves amidst the global competitiveness prevalent today. Customer service is an integral part of the service quality component, making it a crucial element in setting companies apart.

Services, whether in healthcare or any other sector, are inherently intangible, making them challenging to quantify. Therefore, the need for a tool to measure diverse opinions becomes paramount. Regarding the approach to customer service, achieving positive experiences does not adhere to a specific and concrete method, as every customer is unique. However, key factors contributing to successful service include the ability to listen, pay attention, respond courteously, and comprehend behaviors and opinions.

1.5.1 Existing Technologies

Primary Health Care is considered of crucial importance in the Portuguese health system (SNS). International experience demonstrates that the organizational and management models of PHC, as well as the adopted financing strategies and the manner of articulation with other levels of care, can either enhance or compromise the fulfillment of the fundamental principles of health systems, as well as the goals set by health policy in terms of equity, access, efficiency, and quality of care [2].

However, after conducting research, it was found that the National Health Service in Portugal currently lacks a digital health unit evaluation tool, and evaluations are based on paper questionnaires within the health units themselves. With the absence of similar applications being used nationally, there arose a need to study similar techniques employed in other countries. The implementation of a Chinese primary care evaluation tool will be discussed next: data quality and measurement properties.

In 2001, the World Health Organization proposed a global goal of achieving universal primary care in six domains, as established by the Alma-Ata Declaration of 1978: first contact, longitudinally, comprehensiveness, coordination, and person or family-centeredness, and community orientation. These attributes have proven effective in identifying the breadth of primary care services and monitoring the quality of primary care in various countries [3, 4].

Initially, a literature review was conducted to identify instruments measuring the quality of primary care from the perspective of patients. After applying various evaluation criteria (e.g., rigorous development and validation processes, wide endorsement, application in different cultural contexts), four potential instruments were retrieved: EUROPEP, General Practice Assessment Survey/Questionnaire, Primary Care Assessment Survey, and Primary Care Assessment Tool (PCAT) [5]. Subsequently, comparisons of the development, validation, and application of these instruments were carried out [6]. The PCAT was selected as the adopted tool base because it evaluates the performance of facilities and systems from multiple perspectives, closely parallel to the World Health Organization's definition of primary care widely accepted in developing countries and is theoretically grounded and scientifically validated [7].

Analyses were conducted using Stata 9 software. First, different methods of imputing missing data were evaluated. Second, principal component analysis was used to identify domains of primary care, and third, internal consistency and correlations between items and scales were tested. All analyses employed the Likert scale as interval data [8].

1.5.2 Similar Technologies

In this section, we will discuss tools for assessing the quality of services provided by multinational companies operating in Portugal across various sectors unrelated to healthcare.

IKEA

IKEA is a multinational company of Swedish origin, headquartered in the Netherlands, dedicated to the production and retail sale of flat-pack furniture, mattresses, appliances, and food. It was founded in the province of Småland, Sweden, in 1943 by Ingvar Kamprad. Since 2008, the company has held the position of the world's largest furniture retailer [9].

IKEA has developed its own tool for assessing the services it provides. All stores are equipped with tablets typically located near the exits/entrances. The questionnaires are relatively simple, easy to read, and quick to complete, aiming to encourage as many customers as possible to fill them out. The assessment collection system primarily consists of multiple-choice options, avoiding text box responses to prevent user abandonment.

Given that IKEA stores in Portugal offer various services and types of products, their questionnaires enable users to select the type or types of services they utilized during their visit. This is done with the goal of maximizing information filtering. As a result, users only respond to questions related to their visit to the store.

McDonald's

McDonald's Corporation is the world's largest fast-food hamburger restaurant chain, serving approximately 68 million customers per day in 119 countries through 37,000 outlets. Headquartered in the United States, the company originated in 1940 as a barbecue restaurant operated by Richard and Maurice McDonald [10].

McDonald's also possesses its own service evaluation tool, notable for the method employed to encourage customers to complete the questionnaire. Whenever a customer consumes a specific product at one of the chain's restaurants, a QR code and a code composed of characters and numbers are provided at the end of the receipt, accompanied by the message "Share your opinion with us and receive a coupon with an offer." In exchange for completing the survey, the company offers products (usually ice cream) on the customer's next visit.

Following an analysis of similar technologies, it is evident that Portugal currently lacks any form of digital healthcare service evaluation tool. The implementation of such a tool in municipalities will play a crucial role in enhancing healthcare services overall.

1.5.3 Paper Questionnaires vs Digital Questionnaires

Throughout history, paper surveys have been widely favored for information collection. However, in recent decades, a decline in response rates has been observed, coupled with an increase in technological development and internet usage. These changes have propelled the rise and adoption of online surveys [11, 12].

The online questionnaire surpasses the paper format, extending geographical reach, providing greater convenience for researchers and respondents, potentially leading to higher response rates [13].

Online surveys provide the advantage of automating data collection and analysis. With suitable technological features, it's possible to program data collection, eliminate errors, and conduct efficient preliminary analyses, streamlining the research process [14]. This contributes to speeding up research, saving time and resources.

2 Research Methodologies

2.1 Methodologies and Methods Used in the Study

One of the methodologies adopted for the development of this project will be User-Centered Design (UCD). UCD emerged from Human-Computer Interaction (HCI) and is a software design methodology utilized by software developers and designers, enabling the creation of projects that meet users' needs [15].

User-Centered Design is an approach that keeps the user at the center of decision-making throughout the entire project lifecycle. The objective is to ensure that the design of the evaluation tool remains focused on the user.

Understanding the needs of the audience is crucial in project development. It is essential to consistently consider questions such as: Who is the user? What are the key functions they require? What is the most intuitive approach?

Usability testing is vital during the project's development phase. Various checks should be conducted before the final launch to ensure everything is functioning correctly. Therefore, it is always beneficial to put oneself in the client's shoes to be certain that everything is in proper working order.

Usability - The navigability of an application is another crucial aspect to maintain customer satisfaction. Providing clear pathways between different windows in which the user is engaged is essential. Usability is one of the primary factors in the development of this project and will be supported based on the User-Centered Design (UCD) methodology. This methodology allows the exploration of various aspects, such as User Experience (UX), which summarizes the experience obtained from using a software product, User Interface (UI), involving the design of the user interface, icon development, prototyping, heuristic evaluation, usability testing, and other relevant techniques for project development.

Consistency - An important factor in helping the user maintain a sense of spatial orientation is consistency. This is a feature that unifies form and function throughout the customer's experience.

Simple and natural dialogue - The main application of a system should be incorporated in a way that enables intuitive dialogue with the user.

Appropriate feedback - The consumer should be aware that actions have been successfully executed. This can be demonstrated through gratitude messages, changes in the appearance of the page, and animated effects on action buttons, for example.

Reduction of user mental effort - Minimizing difficulty in interaction prevents frustration. Therefore, offering an intuitive design and, if necessary, providing instructions for use should be established from the outset.

Ensuring trust - Transparency in explanations helps create a climate of credibility in favor of the tool to be developed.

2.2 Technology Assessment Models

The concept of technology adoption refers to a complex process that encompasses individuals, technology, and the context of implementation. In this subsection, we will present some of the most reliable technology acceptance models, such as the Technology Acceptance Model (TAM) and its versions TAM2 and TAM3 and discuss the differences among these models.

The TAM was proposed by Davis in 1986 in his doctoral thesis. Since then, it has been tested and extended by numerous researchers. Overall, the TAM has been successfully validated in predicting approximately 40% of system usage [16]. One of the main objectives of the TAM is to provide a foundation for tracking the impact of external variables on internal beliefs, attitudes, and intentions. This suggests that perceived ease of use (PEOU) and perceived usefulness (PU) are the two most crucial constructs for explaining system usage. The TAM was initially adapted to the context of Information Systems (IS) and designed to predict the acceptance and use of information technology

in the workplace. However, subsequently, the TAM has been widely applied to a diverse range of technologies and user contexts [17].

TAM2 is an extended version of the TAM, incorporating additional predictors by Venkatesh and Davis in 2000. TAM2 reflects the impacts of three interrelated social forces that affect an individual's intention to adopt or reject a new or advanced technology. A broad range of moderators, such as subjective norm, voluntariness, age, gender, culture, and image, can be considered in this modeling approach [18].

TAM3 presents a complete homological network of determinants for the adoption and use of IT by individuals. This model suggests three theoretical extensions beyond TAM2 and the model of perceived ease of use determinants [19].

2.3 Data Collection Techniques

The techniques under investigation, to be explored and adopted for data collection, will focus on the foundation of designing and analyzing questionnaires. These questionnaires will adhere to the following models: System Usability Scale (SUS), User Experience Questionnaire (UEQ), Questionnaire for User Interface Satisfaction (QUIS), and Likert scale. These instruments aim to gather data pertaining to the usability of the project under development and the level of acceptance it will receive from users.

2.4 Usability Tests and Evaluations

To measure the usability aspects of the system and identify specific issues, it is essential to plan, develop, and conduct usability tests. The user plays a crucial role in the overall process of interface design development, which involves utilizing the aforementioned techniques to measure usability aspects and engage in iterative cycles of design, prototyping, and evaluation.

Some of the already highlighted but significant points in conducting usability tests will take into account the following factors:

- Learning time required;
- Performance speed and resource availability from the developed system;
- Error rate (in interface usage);
- Measurement of user satisfaction in using the system.

3 Implementation

The health service quality assessment tool for municipalities will be implemented as a web application and subdivided into 5 different layers (presentation layer, business layer, data layer, security layer, and infrastructure layer).

Presentation Layer (Front-end) – The presentation layer will be implemented using the Angular front-end framework. The following languages will be utilized in creating the web application: CSS, HTML, and TypeScript, along with the integration of various web code libraries.

Business Layer (Back-end) – The business layer will be implemented in the C# language, utilizing the Visual Studio software and the ASP.NET Core framework. This is where business rules, data validation, and data processing will be implemented.

Data Layer (Database) – The data layer will be implemented using a relational database in PostgreSQL through the PgAdmin platform. PostgreSQL is a powerful open-source database system with over 30 years of active development, ensuring a strong reputation for reliability, robust feature set, and performance.

Security Layer – The entire security system will be based on JWT Authentication, a data transfer system that can be securely sent via URL, POST method, or in an HTTP header.

Infrastructure Layer – The infrastructure supporting the health service quality assessment application for municipalities is distributed across different cloud environments to optimize performance, scalability, and availability. The database is hosted on Amazon RDS (Relational Database Service), benefiting from the robustness and reliability of AWS (Amazon Web Services) database services. The API, responsible for business logic, is hosted on the Microsoft Azure platform, ensuring effective and efficient implementation. As for the web application for questionnaires and dashboards, both are also implemented on Microsoft Azure, providing a cohesive integration, and facilitating centralized management.

3.1 Application of Questionnaires

This chapter delves into the implementation of the questionnaire application, a crucial component of the project at hand. Accessibility has been a central consideration, with the application easily accessed by scanning a QR code on mobile devices. The adopted design is characterized by simplicity and intuitiveness, aiming to cater to all age groups within the population (Fig. 1). To promote inclusive participation, features such as the ability to zoom in have been incorporated, facilitating question readability for individuals with reduced visibility. Additionally, inclusion is underscored by allowing questions to be read aloud, benefiting individuals with visual impairments. Navigation between questions and the ability to modify selected answers have been implemented to provide a comprehensive and adaptable experience. This approach reflects the project's commitment to accessibility and equitable participation.

3.2 Application of Data Processing and Analysis

This chapter explores the implementation of the application dedicated to collecting responses through questionnaires, a crucial phase of the project. Developed in Angular, the application provides a robust approach for the analysis and processing of collected data. The analysis section is structured into five key points: demographics (Fig. 2), facilities, service, quality, and overall appreciation, offering a comprehensive assessment of municipal health services.

To optimize the utility of the tool, filters have been incorporated allowing users to select the analysis location, date, and a specific set of questions. This flexibility enables personalized and targeted analysis, contributing to a more profound and refined understanding of the various aspects being evaluated. By highlighting the application of advanced technologies and efficient customization of analysis, this chapter illustrates the project's commitment to providing a robust and adaptable tool for the comprehensive evaluation of municipal health services.

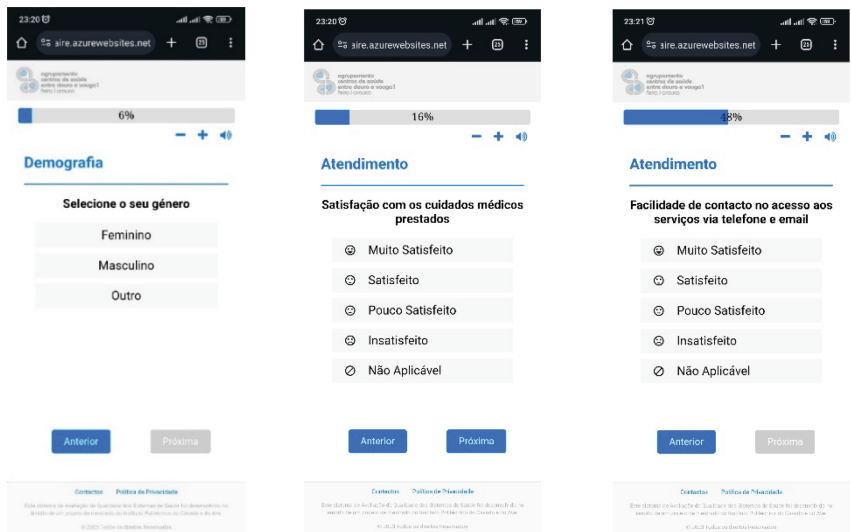


Fig. 1. Layout of the application for collecting responses through questionnaires.

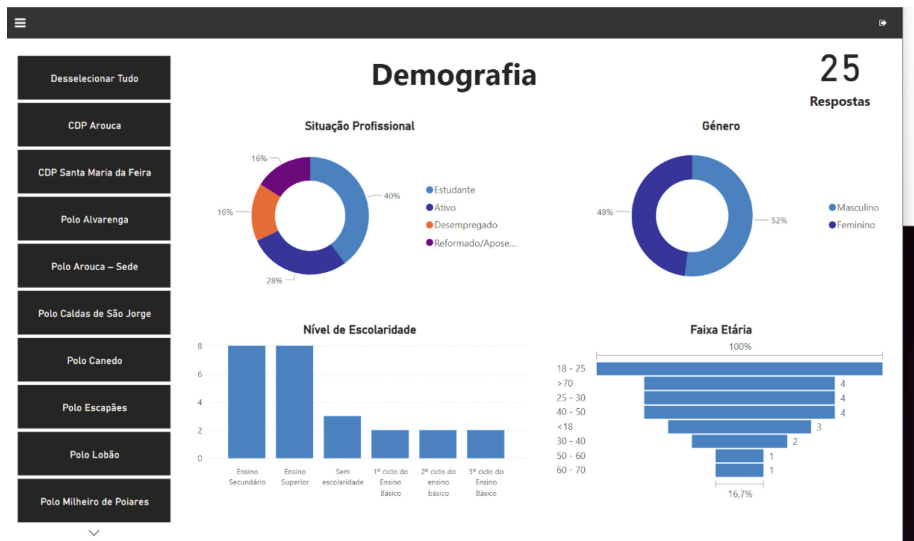


Fig. 2. Layout of the application for collecting responses through questionnaires: demographics.

In addition to the core functionality of response collection, a user management page has been integrated to enhance the efficiency and control of the platform (Fig. 3). In this section, administrators can edit user data, delete user accounts, adjust permission profiles, add new users, and temporarily block a user’s access to the application. This strategic extension aims to provide a flexible and effective administration of the platform, ensuring administrators have the necessary tools to manage user access and participation

efficiently. With this approach, the application not only excels in robust data collection but also in the ability to dynamically administer and adapt the user experience.

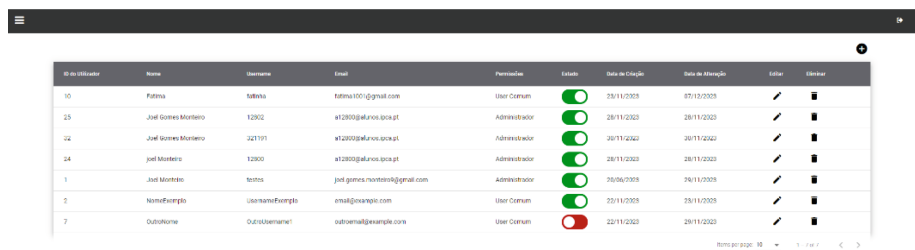


Fig. 3. Layout of the application for collecting responses through questionnaires: user management.

Further enhancing the platform’s functionality, a new page has been added to facilitate the printing of a new QR code (Fig. 4). In this section, the system allows users to select their desired clinic and generate the corresponding QR code for easy access to the application. This strategic addition aims to provide convenience and flexibility, enabling users to easily obtain an updated QR code for interacting with the application in different contexts and locations. The ability to print personalized QR codes represents another step forward in the platform’s ongoing commitment to meeting users’ dynamic needs, making the user experience even more adaptable and efficient.



Fig. 4. Layout of the application for collecting responses through questionnaires: print QR code

4 Final Comments

The project has reached its final phase, gearing up for real-world testing, marking a culmination after months of dedicated development. Following this crucial stage, a comprehensive study on the usability of the applications is scheduled, aiming to evaluate effectiveness and user experience.

At present, the focus is on rectifying minor usability issues in the questionnaire data collection application, with efforts directed towards refining the interface to ensure an intuitive and seamless interaction.

Looking ahead, there is identification of areas for continuous improvement. Speed of access to the application emerges as a priority, with efforts directed at optimizing the experience and efficiency. Graphic enhancements are also under consideration to improve aesthetics and visual appeal. The implementation of new functionalities is on the horizon, aiming to further enrich the utility of the platform.

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